Established results and remaining scope

Fixed ordered cylindrical compositions with sufficiently long net twists

Exact tangle reduction

Combine the prescribed twist groups, retaining rotations. The knot and its normalized invariant are unchanged.

Control of the full sum

Uniform bounds and a nonzero contribution near the geometric solution

Proved in this class

Actual knot volume

$$\lim_{N\to\infty} \frac{2\pi}{N} \log |K_N| = \operatorname{Vol}(S^3 \setminus K)$$

The main knot cusp stays unfilled

The same invariant from a pointed quantum A-operator

Data at finitely many fixed colors determine a period and a positive determinant (Gram) ratio.
Both have the proved knot-volume growth; the original peripheral factors are retained.

Remaining scope

Arbitrary short net twists and more general knot families; extending the geometric A-factor theorem beyond circle trees.

The full-sum estimate is proved for the reduced strict presentations.

The output must be non-elementary, and the actual closed link must be a knot.